

KEVIN M. SMITH

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SUMMARY

- Ph.D. Candidate in Environmental and Water Resources Engineering at Tufts University
- Researching semi-autonomous civil infrastructure as a technology for mediating environmental conflicts
- Recipient of the NSF Integrative Graduate Education and Research Traineeship (IGERT) on Water and Diplomacy
- Interests: Water and Diplomacy, Science and Technology Studies, Social Choice, Risk Measures, Human-Robot Interaction

EDUCATION

- Ph.D. Candidate, Environmental and Water Resources Engineering, Tufts University, since 2013 (Advisor: Dr. Shafiqul Islam)
- B.S., Earth and Environmental Engineering, Columbia University, 2013 (Cum Laude, Tau Beta Pi)
- B.A., Environmental Studies, Oberlin College, 2013 (Joyce Gorn Memorial Prize for Research)

REFEREED JOURNAL PUBLICATIONS

- **2026 - From Rainfall to Rumors: Interpreting the 2024 Bangladesh Floods Through Numbers, Narratives, and Information Gaps**
 - P. Das, A. R. Ganguly, N. Rabb, **K. M. Smith**, S. Islam
 - Water Resources Research, doi: 10.1029/2025WRO42679.
- **2024 - Reconstructing Decision-Making Dynamics During Public Health Crises by Applying Data Science to Public Records**
 - P. Nadel, **K. M. Smith**
 - Journal of Public Health Policy, doi: 10.1057/s41271-024-00540-y.
- **2023 - On Exponential Utility and Conditional Value-at-Risk as Risk-Averse Performance Criteria**
 - **K. M. Smith**, M. P. Chapman
 - IEEE Transactions on Control Systems Technology, 1558-0865, doi: 10.1109/TCST.2023.3274843.
- **2022 - On Optimizing the Conditional Value-At-Risk of a Maximum Cost for Risk-Averse Safety Analysis**
 - M. P. Chapman, M. Fauß, **K. M. Smith**
 - IEEE Transactions on Automatic Control, 1558-2523, doi: 10.1109/TAC.2022.3195381.
- **2021 - Risk-Sensitive Safety Analysis Using Conditional Value-at-Risk**
 - M. P. Chapman, R. Bonalli, **K. M. Smith**, I. Yang, M. Pavone, C. J. Tomlin
 - IEEE Transactions on Automatic Control, 1558-2523, doi: 10.1109/TAC.2021.3131149.
- **2021 - Classical Risk-Averse Control for a Finite-Horizon Borel Model**
 - M. P. Chapman, **K. M. Smith**
 - IEEE Control Systems Letters, 2475-1456, doi: 10.1109/LCSYS.2021.3114126.
- **2021 - Addressing Complex Challenges in Coupled Natural and Human Systems Through Principled Pragmatism**
 - **K. M. Smith**, W. Palash, E. Choudhury, S. Islam
 - Frontiers in Water, Volume 3, 2021, doi: 10.3389/frwa.2021.61725.
- **2014 - Forecasting Energy Consumption of Multi-family Residential Buildings Using Support Vector Regression**
 - R. K. Jain, **K. M. Smith**, P. J. Culligan, J. E. Taylor
 - Applied Energy, Volume 123, 2014, pp. 168-178, doi: 10.1016/j.apenergy.2014.02.057.

REFEREED CONFERENCE PUBLICATIONS

- **2019 - A Risk-Sensitive Finite-Time Reachability Approach for Safety of Stochastic Dynamic Systems**
 - M. P. Chapman, J. Lacotte, A. Tamar, D. Lee, **K. M. Smith**, V. Cheng, J. F. Fisac, S. Jha, M. Pavone, C. J. Tomlin
 - 2019 American Control Conference (ACC), pp. 2958-2963, doi: 10.23919/ACC.2019.8815169.
- **2018 - Reachability Analysis as a Design Tool for Stormwater Systems**
 - M. P. Chapman, **K. M. Smith**, V. Cheng, D. L. Freyberg, C. J. Tomlin
 - 2018 IEEE Conference on Technologies for Sustainability (SusTech), pp. 1-8, doi: 10.1109/SusTech.2018.8671362.

EDITED VOLUMES

- **2025 - The Routledge Handbook of Water Diplomacy**
 - Edited by S. Islam, **K. M. Smith**, M. Klimes, A. Salzberg
 - Routledge. London, England. 743 pages. ISBN: 9781032013893
- **2020 - Interdisciplinary Collaboration for Water Diplomacy: A Principled and Pragmatic Approach**
 - Edited by S. Islam, **K. M. Smith**
 - Routledge. Abingdon, Oxon. Earthscan Series in Water Resource Management. 306 pages. ISBN: 9781138369283

SOFTWARE

- **Arduino SDI-12 (C++)** github.com/EnviroDIY/Arduino-SDI-12
 - first open-source library implementing the SDI-12 communication protocol for open-hardware Arduino-based data loggers
 - originally authored in 2013 by **K. M. Smith**, now maintained by S. Damiano.